

Eve: Being a small start-up, the overall data requirements will not exceed a few terabytes of storage. We deal with unstructured data like audio/video files and raw or text based data. Scalability is an important factor, as the solution must be able to handle a multi-fold increase in data as the business expands.

Bob: Keeping your basic requirements in mind, I can provide you with two alternatives. You can go with on-premise solutions or you can opt for cloud based storage. Both the solutions have their own pros and cons.

Eve: So what is the basic difference between the two options? How do I decide which one is better for my organisation? There has been a lot of buzz about choosing the right storage environment. In case I do choose an on-premise option for data storage, I have heard about SAN, NAS and DAS (storage area network, network attached storage and directly attached storage, respectively) solutions? What are the specific features provided by each of these?

Bob: Okay, now let me walk you through a few simple scenarios. Let's start with on-premise solutions, with which all the components remain under your own supervision at a local or remote site, under your control.

Let's take an example since yours is a start-up, it's advisable to build a computer/server with an array of storage disks, let's say about 5-10TB of hard drive space. These additional hard drives would act as DAS, which is a simple storage system like a hard disk or any block device connected to the computer/server. DAS can be daisy-chained if you plan to spend some additional money later to expand your storage. So Eve, for your business, DAS is preferred due to the following reasons—lower latency and costs, as well as higher speed when compared to directly accessing raw data. DAS is also easy to configure and deploy when compared to SAN or NAS. It is the preferred solution for databases, and for file and disk intensive operations.

Eve: Adding more storage, i.e., additional disks to our existing solution would definitely increase storage but what about redundancy or data backup; how do I manage that in the case of DAS?

Bob: I would suggest adding one RAID card to your newly built computer/server and configure the desired RAID.

Eve: That's good Bob, but still, the solution is centralised. What if I want distributed computing and availability over the network?

Bob: In such a scenario, you can use your DAS array and share it over the network. This will act as a file server over the network. Unlike SAN and NAS, which require a computer with processor, memory and operating system, DAS comprises hardware devices only. Network attached storage (NAS) provides fast, simple, reliable access to data in an IP networking environment. Though NAS is good for collaboration and sharing, it's not ideal in case of mission critical applications like heavily used databases and since it relies on the network, that would add to the latency.

Eve: So then what is SAN? How is it different?

Bob: SAN is an option when your applications require

the performance of DAS and the benefits of NAS. At the data center level, virtual or cloud computing SAN is the perfect solution for high-end computing, but requires complex configuration especially in case of virtual networks, where different switch rules are used. Now coming to the differences, NAS creates a shared folder, which can be accessed by any user over the network and we cannot install the OS on the folder. On the other hand, in the case of SAN, we create logical partitions and share them over the network, which in turn allows us to install the OS on that logical partition. SAN provides higher hardware utilisation and speed and is the best solution for virtualisation and cloud computing scenarios.

Some open source solutions to manage NAS and SAN are OpenFile (<https://www.openfiler.com/>), FreeNAS (<http://www.freenas.org>) and Gluster (<http://www.gluster.org/>).

Eve: But with the ever increasing complexity and variety of data, the rising need for zero down time applications, etc., can these traditional solutions still provide the same efficiency and functional advantages?

Bob: Though these legacy approaches are still prevalent, there has been a lot of innovation in the field of storage technology in the past decade. As you might have noticed, in all the traditional solutions, the management functionality is provided or built-in at the hardware level itself.

Software-defined storage delivers automated, policy-driven, application-aware storage services by amalgamating the underlying storage infrastructure in support of an overall software defined environment. Open Source Software Defined Storage (OpenSDS) enables companies to achieve massive, virtually limitless scalability even while using the commodity hardware, thereby, virtually eliminating the vast silos of distributed stored data behind traditional arrays and provides commodity based solutions. Few examples of OpenSDS are Skylable (<http://www.skylable.com/>) and StorMax SDS (<http://www.amax.com/enterprise/sds.asp>).

Eve: That's interesting, but are there any downsides to SDS?

Bob: A few of the limitations of SDS are that there are more components to manage (due to the modular architecture), less hardened solutions due to limited innovation on this front, and the need for more infrastructure to achieve a high performance that's comparable to existing distributed, storage arrays.

Eve: So are there any innovative solutions that overcome these drawbacks?

Bob: In SDS, where the compute, network and storage are treated as separate layers, there is a management overhead induced. So, an alternative approach called converged storage systems that consolidates all these into one layer to simplify management comes into the picture. Converged storage systems provide one resource pool solution and hence make deployment faster and lower administration costs. With a virtualised storage layer, it is easier to use such systems in a virtualisation platform. With these systems, overall resource utilisation is higher